Manifold LLC

Independent AI/ML advice and engineering for Intelligence Community and Department of Defense programs

Manifold LLC is a Virginia small business run by two cleared practitioners with about ten years each inside IC acquisition and research programs. We help program offices decide whether to build or buy, and how to get the result running on accredited networks. When the answer is build, we build; never for an organization we also advise.

CORE COMPETENCIES

Strategy and roadmaps

Foundation-model strategy, roadmaps, build-versus-buy analysis, technology forecasts. Technical SME on competitive acquisitions and BAA panels; AI/ML training for sponsor staff.

Acquisition and compute

Technical requirements, RFP and SOW language, source-selection criteria, and vendor evaluation. GPU and TPU cluster sizing and cost models across AWS GovCloud, Azure Government, and Google Cloud's assured regions, or on-premises hardware.

Program assessment

Independent technical evaluation of AI/ML programs, maturity and risk assessment, milestone and design reviews, sponsor-facing metrics.

Secure generative AI

Reference architectures for CUI at IL4 and IL5, retrieval-augmented generation design and delivery, model and vendor selection in government cloud regions, evaluation and bias audits.

Applied research and development

Detectors and trackers over satellite imagery, foundation models pretrained on sponsor data with scaling studies, CPU, GPU and TPU implementations priced per run. Evaluation protocol first; negative results reported.

DIFFERENTIATORS

- Both principals hold TS/SCI with CI polygraph
- Founding principals deliver the work themselves
- Hands-on with both acquisition and model training
- Delivered production generative AI at IL4 on GovCloud
- Advise or build, never both for one client; no vendor fees

COMPANY DATA

LEGAL NAME	Manifold LLC (Virginia, formed 2026)
LOCATION	Loudoun County, Virginia
SIZE	Small business
UEI	C6P2G6PQLEB8
CAGE	23X60
NAICS	541715, 541512, 541611, 541690
CLEARANCES	TS/SCI with CI polygraph, both principals
CLASSIFIED WORK	On customer premises
WORK TYPES	Direct award, subcontract, teaming
PRINCIPALS	Paul G. Singerman, Ph.D.; John "Luke" Och

POINT OF CONTACT

CONTACT	Paul G. Singerman, Principal
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RELEVANT EXPERIENCE OF THE PRINCIPALS

MARITIME DOMAIN AWARENESS · RESEARCH

Built and measured a tracker that re-finds ships that have switched off AIS from one satellite pass to the next, over the waters around Denmark and southern Norway, using public radar, optical and night-light sensors (Sentinel-1, NISAR, Sentinel-2, Landsat, VIIRS), a motion model learned from historical AIS, and scoring against withheld Danish and Norwegian AIS. No published system was found that measures this problem; the results stand as a first public baseline. Unpublished draft.

SYNTHETIC APERTURE RADAR · OPEN-SOURCE SOFTWARE

FastSAR, a library that forms SAR images on CPUs, NVIDIA GPUs and Google TPUs through one Python interface (exact and factorized backprojection, polar format; CPHD, SICD, Sentinel-1 and NISAR inputs). Every configuration scored against a high-precision reference and priced at cloud rates: faster than every open-source implementation tested, by orders of magnitude in most cases, at equal or better accuracy. Open source (MIT) at github.com/saulpingerman/FastSAR and on PyPI as fastsar; preprint arXiv:2610.33932, submitted to IEEE TGRS.

VESSEL MOVEMENT · FOUNDATION MODEL

A single model pretrained on public AIS traffic that forecasts a vessel's position as a probability map at any horizon, with small task heads on the frozen model for analysts' questions. Paper under review.

NAVY UARC · GENERATIVE AI

Designed and delivered a production retrieval-augmented generation system for CUI on Amazon Bedrock in an Impact Level 4 AWS GovCloud environment, integrated with the laboratory's directory service (LDAP).

GEOSPATIAL FOUNDATION MODELS

Architected the modeling and training effort for foundation models over overhead and geospatial data, working directly with vendor engineering teams (Google, NVIDIA) on data pipelines, model design and compute planning.

INTELLIGENCE COMMUNITY ACQUISITION

About ten years inside IC acquisition offices and research programs, one principal as a Navy UARC trusted agent and the other as a SETA lead: technical requirements, source-selection and BAA evaluation, AI/ML training design, and senior-leadership briefings.

Some of this work predates the firm. Details and references are available in conversation.

PRINCIPALS

Paul G. Singerman, Ph.D.

Associate Research Professor at Penn State's Applied Research Laboratory, a Navy University Affiliated Research Center, where he leads the laboratory's largest AI/ML research program: more than thirty engineers, over ten million dollars a year, and a sole-source follow-on contract with a ceiling above two hundred million dollars. Since 2023 the laboratory's on-site trusted agent to an Intelligence Community agency: contributed to its five-year foundation-model roadmap, technical evaluator on five competitive acquisitions and on BAA panels, coordinated an agency-wide AI/ML training curriculum. Research in geospatial foundation models, SAR image formation and data fusion. Ph.D. electrical engineering and M.A. mathematics, Penn State. TS/SCI with CI polygraph.

John "Luke" Och

Leads a team of ten systems engineering and technical assistance (SETA) engineers at SAIC, supporting early-stage AI/ML initiatives for a national security space program: acquisition strategies, technical requirements for more than ten low-maturity AI/ML capabilities, and the performance metrics and evaluation criteria behind investment decisions. Five years before that as a satellite ground systems engineer, running source selections from RFP development through proposal evaluation and contract negotiation. Regents Bachelor of Arts, West Virginia University (mathematics and communications); SAFe Agilist. TS/SCI with CI polygraph.